

Answer **all** questions in the spaces provided.

1. (a) State, in terms of electrons, the difference between ionic and covalent bonding. [2]

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- (b) When magnesium reacts with oxygen, the ionic compound magnesium oxide is formed.

- (i) The outer electrons of magnesium and oxygen are shown below. Draw arrows to show the transfer of electrons when magnesium oxide is formed. [1]



- (ii) State the charges on the ions formed [2]

Mg O

